

Multimedia Data Compression

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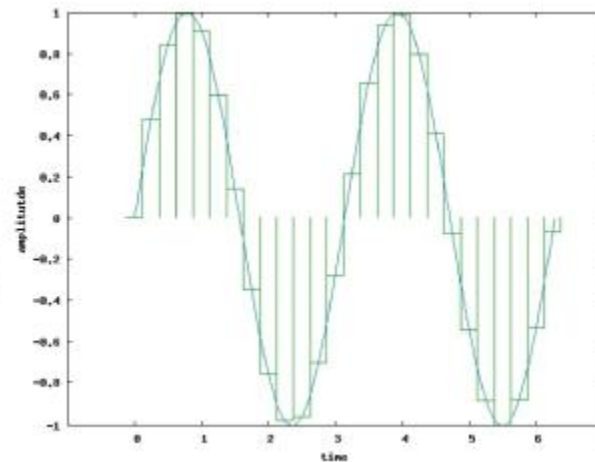
- Audio compression
- Image compression
- Video compression
- Text compression

Lossless vs Lossy Compression

- Lossless: original data recovered exactly — required for text, medical/legal archives, PNG
- Lossy: discards imperceptible or redundant information — used for audio (MP3), image (JPEG), video (H.264)
- Compression ratio = original size / compressed size (e.g., 10:1 means 90% reduction)
- Bitrate (audio/video): bits per second — lower bitrate → more compression, potential quality loss
- Trade-off triangle: compression ratio $\leftrightarrow$ quality $\leftrightarrow$ computational cost
- Multimedia DB stores both compressed BLOBs and metadata (codec, bitrate, quality level)

Audio compression

- Audio signals are continuous in both time and amplitude.
- Sound waves are converted into electrical signals
- The electrical signals are then sampled, quantized, and encoded.
 - It is necessary to determine the sampling frequency, quantization levels, and related parameters,...



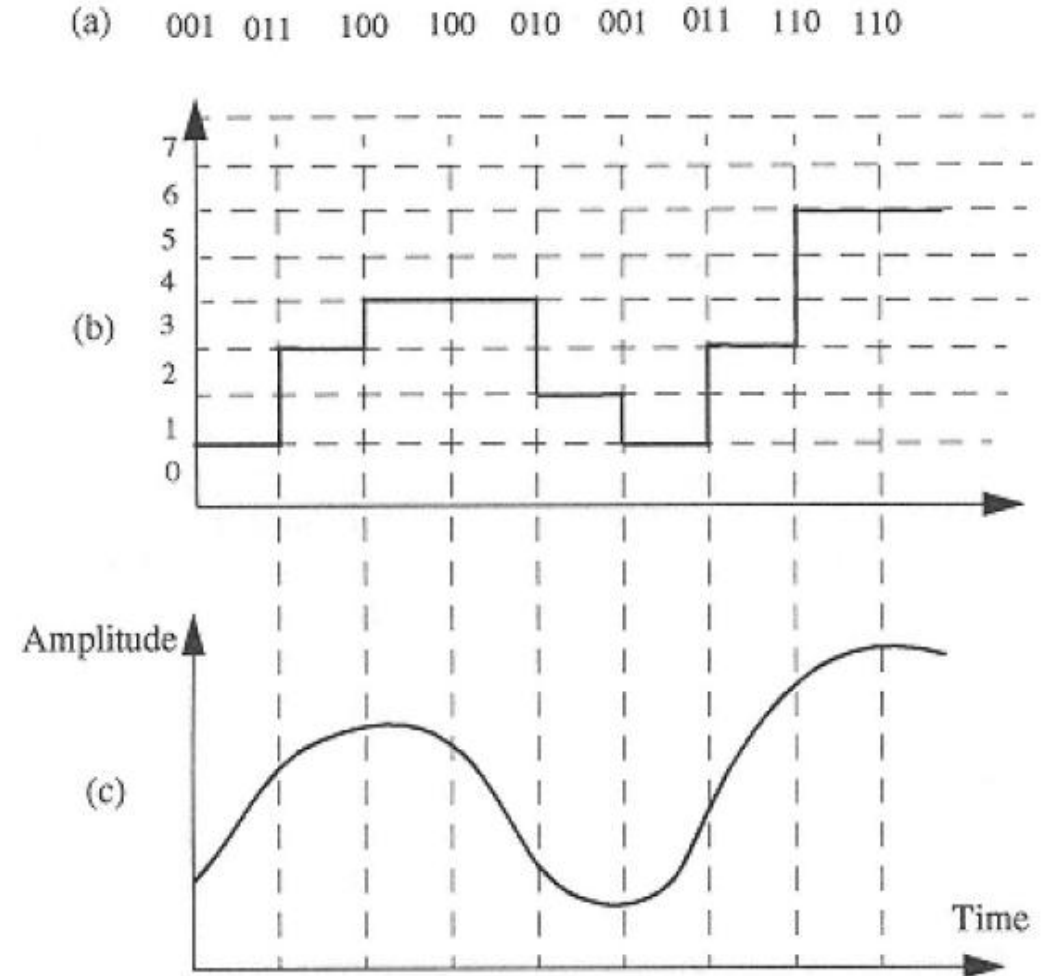
Sample values

0.0000000000000000
0.479425538604203
0.841470984807897
0.997494986604054
0.909297426825682
0.598472144103956
0.141120008059867
-0.35078322768962
-0.756802495307928
-0.977530117665097
-0.958924274663138

...

Audio compression

- Disadvantages of linear quantization include:
 - Imbalance in the signal-to-noise ratio (SNR) between high-amplitude and low-amplitude regions
 - The human auditory system is more sensitive to sounds in low-amplitude regions



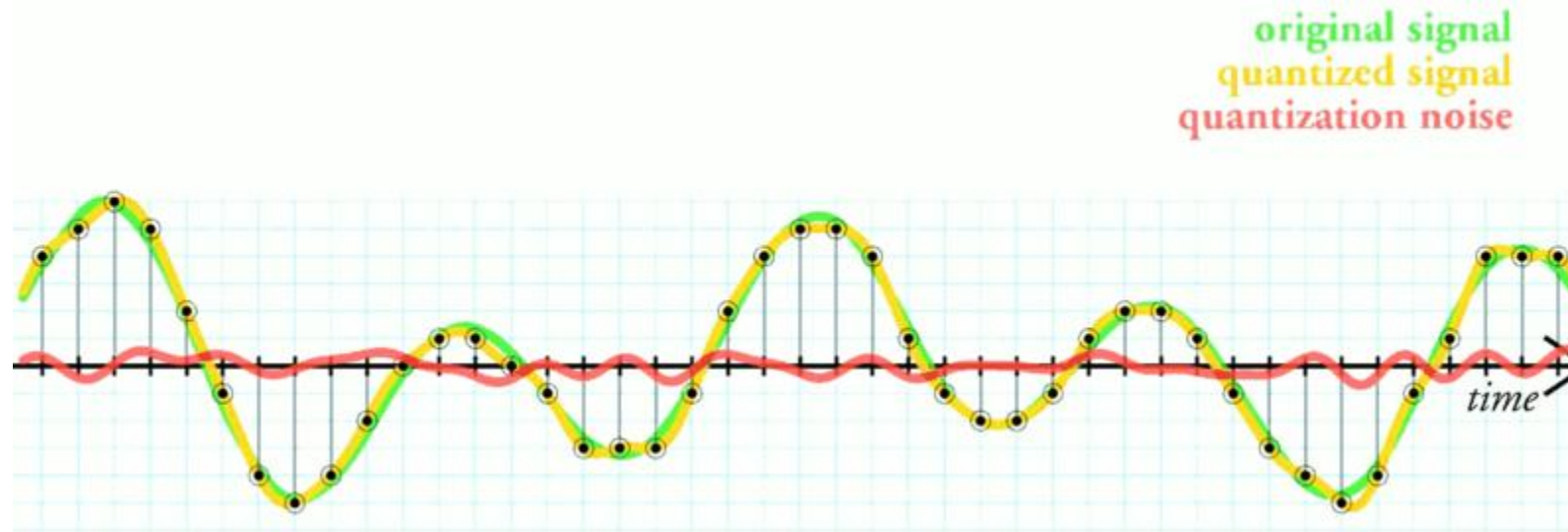
Exercise 2 — Sampling and Quantization

- A telephone channel carries speech with maximum frequency 3.4 kHz.
 - (a) What is the minimum Nyquist sampling rate?
 - (b) If sampled at 8 kHz with 8-bit linear quantization, what is the bit rate?
 - (c) Why does ITU-T G.711 use nonlinear (A-law) quantization instead of linear 8-bit?
 - (d) Compare SNR impact on quiet vs loud speech segments under linear quantization

PCM Sampling and Quantization

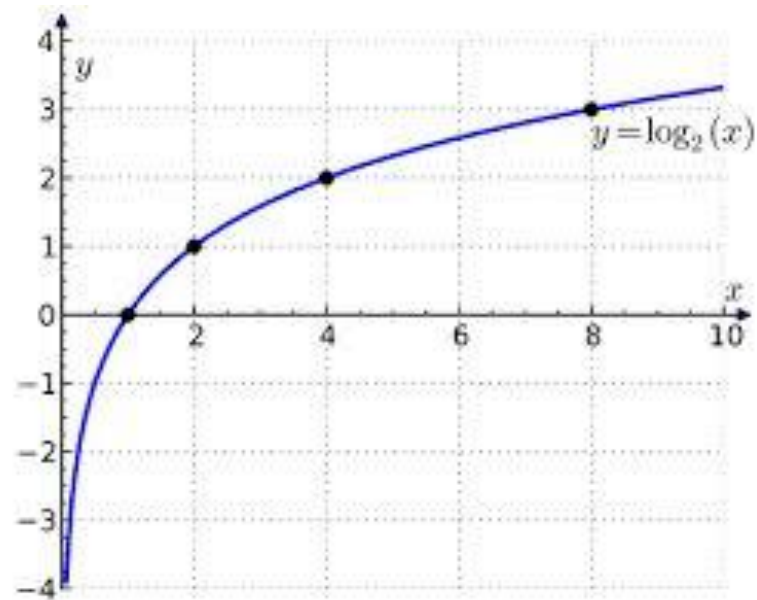
- Sampling: convert continuous audio to discrete samples at rate f_s (Hz)
- Nyquist theorem: $f_s \geq 2 \times f_{\max}$ to avoid aliasing (speech ~ 8 kHz $\rightarrow f_s \geq 16$ kHz; audio CD = 44.1 kHz)
- Quantization: map each sample amplitude to nearest of 2^B levels (B = bit depth; CD = 16-bit = 65,536 levels)
- Bit rate (PCM) = $f_s \times B \times \text{channels}$ (e.g., $44100 \times 16 \times 2 = 1,411,200$ b/s ≈ 1411 kb/s)
- Linear quantization: equal step size — poor SNR at low amplitudes
- Leads to nonlinear quantization (μ -law, A-law) and perceptual codecs (MPEG Audio)

Audio compression



Audio compression

- Nonlinear quantization:
 - The quantization step size varies logarithmically with the amplitude of the audio signal
 - The step size is small when the signal amplitude is low and increases as the amplitude becomes larger.



Audio compression

- Predictive coding
 - The amplitude value of the signal is predicted
 - The error (difference) between the predicted value and the actual value is computed
 - This error is then quantized and encoded
 - Differential Pulse-Coded Modulation (DPCM)
 - Fixed-length codes are used in DPCM

Audio compression

- DPCM is effective only when
 - Adjacent amplitude values of the signal are correlated
 - The prediction error is smaller than the range of the actual signal amplitudes, allowing fewer bits to be used to represent the error

Audio compression

- Adaptive DPCM
 - This approach is applied to signals whose amplitudes vary either very rapidly or very slowly.
 - When the signal amplitude changes rapidly, a large quantization step size is used
 - When the signal amplitude changes slowly, a small quantization step size is used.

DPCM — Worked Example

- Previous sample $x[n-1] = 200$. Current sample $x[n] = 203$. Predictor: $\hat{x}[n] = x[n-1]$
- Prediction error $e[n] = x[n] - \hat{x}[n] = 3$ (much smaller range than full 0–255 amplitude)
- Quantize error $e[n]$ with fewer bits $\rightarrow$ transmit code for $e[n]$; decoder reconstructs $x[n]$
- Savings: if amplitudes need 8 bits but errors need only 3 bits $\rightarrow$ ~62% reduction on samples
- Works when consecutive samples are correlated (speech, smooth music)
- Fails when signal changes abruptly (transients, percussion) — ADPCM adapts step size

Exercise — DPCM and ADPCM

- Sample sequence: 100, 102, 101, 103, 104, 140, 144 (8-bit audio)
 - (a) Using predictor $\hat{x}[n] = x[n-1]$, compute prediction errors for $n = 2..7$
 - (b) What is the max error magnitude before sample 140? After sample 140?
 - (c) Why does ADPCM increase step size after sample 140?
 - (d) Compare fixed DPCM vs ADPCM suitability for this signal

Audio compression

- Recommendations of the International Telecommunication Union – Telecommunication Standardization Sector (ITU-T) for speech signal coding (with bandwidths of 3.4 kHz or 7 kHz)

<i>Recommendation</i>	<i>Compression technique</i>	<i>Speech bandwidth (kHz)</i>	<i>Sampling rate (kHz)</i>	<i>Compressed bit rate (kbps)</i>
G.711	PCM (no compression)	3.4	8	64
G.721	ADPCM	3.4	8	32
G.722	Subband ADPCM	7	16 (14 bits per sample)	48,56,64
G.723	ADPCM	3.4	8	24
G.728	Vector quantization	3.4	8	16

Audio compression

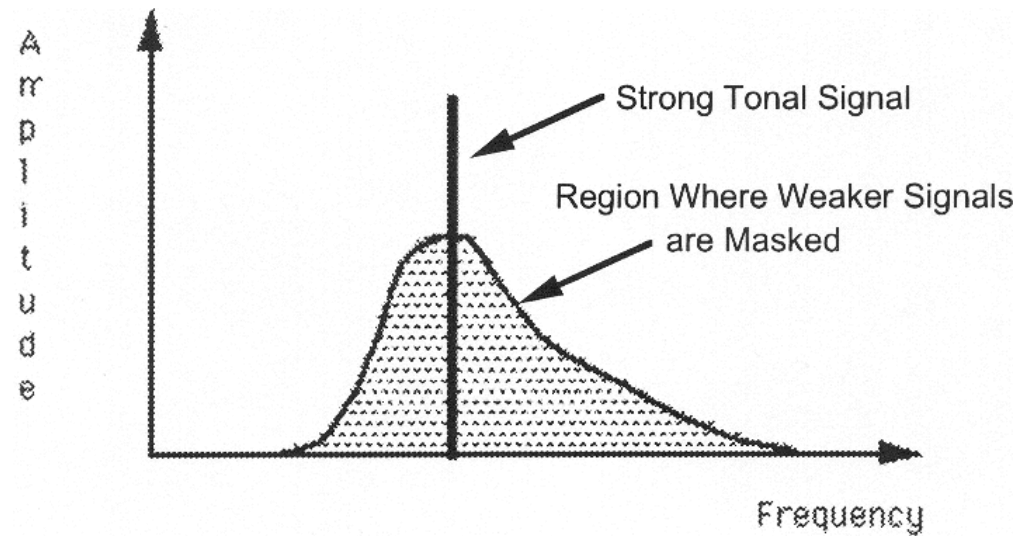
- MPEG-Audio: masking effects
 - Audio signal coding over the full audible frequency range (20 Hz–20 kHz)
 - Masking properties: one sound can mask (render inaudible) another sound, or it can alter the perceived intensity of another sound.
 - If a sound is masked, it can be removed without affecting the perceived signal quality
 - Sampling frequencies: 32, 44.1, 48 kHz.

Psychoacoustic Model and Frequency Masking

- Human hearing is limited to ~ 20 Hz–20 kHz; sensitivity varies by frequency (Fletcher-Munson curves)
- Frequency masking: a loud tone raises the hearing threshold for nearby frequencies
- Temporal masking: a loud sound masks quieter sounds shortly before/after it
- MPEG Audio uses a psychoacoustic model to compute masking threshold per frequency sub-band
- Inaudible components (below masking threshold) are allocated fewer bits or discarded
- Result: perceptually transparent compression at $\sim 10:1$ vs PCM (128 kb/s MP3 vs 1411 kb/s CD)

Audio compression

- The auditory masking threshold
 - Depends on the sound frequency
 - Depends on the energy of the sound wave within a certain bandwidth around a selected frequency.



Audio compression

- Basic steps:
 - Convert the signal to the frequency domain
 - Partition the signal into two components: audible and inaudible
 - Determine the minimum masking threshold for each frequency band
 - Compute the ratio between the selected signal components and the masking signal.

Audio compression

- MPEG-audio: It allows audio encoding at three levels
 - MPEG-audio layer 1
 - MPEG-audio layer 2
 - MPEG-audio layer 3

MPEG Audio Layers — Comparison

- Layer I: simplest filter bank (32 sub-bands), frame = 384 samples, bit rate $\geq$ 128 kb/s/ch
- Layer II: finer scale factors, frame = 1152 samples, used in DAB and Video CD
- Layer III (MP3): MDCT + hybrid filter bank, Huffman coding of spectra, best compression
- All layers: same psychoacoustic model concept; Layer III adds entropy coding and bit reservoir
- Typical bit rates: Layer I $\sim$ 384 kb/s stereo; Layer II $\sim$ 256 kb/s; Layer III 128–320 kb/s
- MP3 (Layer III) dominates file sharing; AAC (MPEG-4) is successor for streaming

Audio compression

- MPEG-audio layer 1:
 - It has the lowest computational complexity
 - It supports bit rates greater than 128 kb/s per channel
 - Each signal frame consists of a header, a CRC check word (optional), and auxiliary data (optional).
 - Example: Philips Digital Compact Cassette.

Audio compression

- MPEG-audio layer 2
 - It has moderate computational complexity
 - It operates at a bit rate of 128 kb/s per channel
 - It is used in Digital Audio Broadcasting (DAB)
 - It supports audio–video synchronization in CD-ROM applications
 - Each signal frame contains 1,125 samples per transmission channel

Audio compression

- MPEG-audio layer 3:
 - It is the most complex
 - It provides the highest audio quality
 - The transmission bit rate is 64 kb/s per channel
 - It is suitable for audio transmission over ISDN networks

Audio compression

- The development of MPEG-audio:
 - MPEG-1: Video CD and MP3.
 - MPEG-2: Digital Television set top boxes and DVD
 - MPEG-4: Fixed and mobile web
 - MPEG-7: description and search of audio and visual content
 - MPEG-21: Multimedia Framework

Image compression

- A digital image is characterized by multiple attributes
 - Image size: width $\times$ height ($x \times y$)
 - Color resolution (number of bits used to encode color): b
- Image data size = $x \times y \times b$.
- Image compression refers to reducing the image data size



(a) JPEG low compression



(b) JPEG high compression

Image Data Size and Redundancy

- Image data size (uncompressed) = width × height × bits per pixel (bpp)
 - Example: 1920×1080 RGB (24 bpp) = $1920 \times 1080 \times 24 = 49,766,400$ bits ≈ 5.93 MB
- Spatial redundancy: neighboring pixels correlate (same object, smooth regions)
- Spectral redundancy: human eye less sensitive to high-frequency detail and color (chroma)
- Psychovisual redundancy: discard information below visual perception threshold
- Coding redundancy: use variable-length codes for frequent symbols (Huffman)

Image compression

- Fundamental principles applied in image compression include:
 - The human visual system has limited ability to distinguish differences between closely spaced pixels
 - Pixels located on adjacent lines are often similar, indicating spatial redundancy.
 - Spatial redundancy can be removed using predictive coding and transform coding techniques.
 - Some image information can be reduced without degrading the informational content of the image.
 - Image characteristics that are not perceptually significant to the human eye can be discarded

Image compression

- Spatial subsampling
 - At the encoder, pixels are selected at certain intervals for transmission.
 - At the decoder, the full set of pixels is reconstructed using interpolation, resulting in an image with lower spatial resolution.
 - For pixels containing both luminance and chrominance components, the chrominance component can be sampled at a lower rate and quantized more coarsely, since the human visual system is less sensitive to color information.
 - Spatially sampled image signals can be further compressed using other coding techniques.

Image compression

- Predictive coding
 - It is fundamentally similar to audio signal compression
 - The color values of neighboring image elements are correlated.
 - The color value of one pixel can be used to predict the color of adjacent pixels
 - One-dimensional prediction exploits the similarity among pixels along the same scan line.
 - Multidimensional prediction additionally utilizes similarities among pixels across different scan lines as well as between different frames.

Image compression

- Transform coding
 - The correlations among pixels are transformed into a set of independent coefficients.
 - These coefficients are transmitted, thereby reducing information redundancy in the image.

Image compression

- The procedure of transform coding is as follows
 - The image is divided into small rectangular blocks (e.g., 8×8)
 - Each image block is transformed
 - Karhunen-Loeve transform (KLT)
 - Discrete cosine transform (DCT)
 - Walsh-Hadamard transform (WHT)
 - Discrete Fourier transform (DFT)
 - The transformed coefficients are selected and quantized for transmission or storage.
 - Huffman coding or run-length coding may subsequently be applied.

DCT and Transform Coding — Intuition

- 8×8 pixel block → 2D DCT → 64 frequency coefficients (DC = average, AC = detail levels)
- Energy compaction: most energy in few low-frequency coefficients (top-left of block)
- Quantization: divide each coefficient by $Q[i, j]$ from quantization table → many become zero
- Zigzag scan: traverse AC coefficients low-to-high frequency → long runs of zeros
- Run-length + Huffman coding on zigzag sequence achieves high compression
- KLT is optimal but data-dependent; DCT is fixed, fast, and near-optimal for natural images

Image compression

- Vector quantization
- Fractal coding.
- Wavelet-based compression.
- In practical systems, the above techniques are combined in various ways.
 - Spatial and temporal sampling.
 - DPCM coding.
 - Two-dimensional DCT-based transform coding.
 - Huffman coding.
 - Run-length coding.

Image compression

- The JPEG (Joint Photographic Experts Group) standard for still image compression
 - It has been widely used since the 1990s.
 - It comprises multiple coding modes.
 - Baseline DCT-based coding (lossy) is the fundamental and most widely used method.
 - Extended DCT-based coding (lossy).
 - Lossless coding.
 - Hierarchical coding, in which images are encoded at multiple resolutions.

Image compression

- Baseline DCT-based coding.
 - The image is divided into small blocks (8×8 pixels).
 - A two-dimensional DCT is applied to each block, producing an 8×8 set of DCT coefficients per block.
 - The coefficients are quantized using a predefined quantization table.
 - The quantized coefficients are scanned in a zigzag order to form a one-dimensional sequence.
 - The first coefficient is referred to as the direct current (DC) coefficient, while the remaining ones are alternating current (AC) coefficients.
 - Huffman coding or run-length coding is then applied to this sequence to compress the data.

Image compression

- JBIG (Joint Bi-level Image Experts Group)
- JPEG-2000

JPEG vs JPEG-2000 and JBIG

- JPEG (1992): DCT-based, block artifacts at low bitrates, lossless mode (JPEG-LS) separate
- JPEG-2000: wavelet transform (DWT), no fixed 8×8 blocks, better at low bitrates, scalable (progressive)
- JBIG: lossless compression for bi-level (1-bit) images — fax, scanned text, line art
- Use JPEG for photos; JPEG-2000 for medical imaging, digital cinema; JBIG for binary documents
- In MMDB: store format tag (image/jpeg, image/jp2) and compression parameters in metadata

Video compression

- Video is a sequence of images
 - There is spatial redundancy.
 - There is temporal redundancy.
- Video compression involves
 - Spatial information compression (similar to image compression).
 - Temporal information compression..

Video Compression — Spatial and Temporal Redundancy

- Raw video bitrate = width × height × bpp × fps (1080p30 RGB: ~1.5 Gb/s — impractical to store)
- Spatial redundancy (I-frames): compress each frame as an image (intra-frame, like JPEG)
- Temporal redundancy (P/B-frames): most pixels unchanged or shifted from previous frame
- GOP structure: I-P-P-P-B-B-P... — I = keyframe, P = forward prediction, B = bidirectional
- Only residuals + motion vectors transmitted for inter-frames → 20:1 to 100:1 compression
- Standards: MPEG-2 (DVD), H.264/AVC (Blu-ray, streaming), H.265/HEVC (4K, 50% smaller than H.264)

Video compression

- Motion estimation and compensation
 - Motion refers to the displacement of pixel positions from a previous frame to a subsequent frame.
 - Each frame is divided into smaller image blocks.
 - Matching blocks between two different frames are identified and represented by motion vectors.
 - The motion vectors, together with the residual (non-matching) pixel blocks, are encoded and transmitted or stored.

Video compression

- MPEG (Moving Picture Experts Group)
 - MPEG-1 encodes VHS-quality video (360×280 pixels, 30 frames per second) at a bit rate of approximately 1.5 Mbps.
 - MPEG-2 encodes video according to the CCIR 601 digital television standard (720×480 pixels, 30 frames per second) at bit rates ranging from 2 to 10 Mbps.
 - MPEG-3
 - MPEG-4
 - MPEG-7

MPEG Video Standards — Overview

- MPEG-1: 352×240 @ 30fps, ~1.5 Mbps — Video CD, early web video
- MPEG-2: 720×480 (SD) / 1920×1080 (HD), 2–15 Mbps — DVD, digital TV, HDTV broadcast
- MPEG-4 Part 2 (DivX/Xvid): object-based coding; Part 10 = H.264/AVC (most widely deployed)
- MPEG-4 Part 10 (H.264/AVC): block motion + DCT/integer transform, CABAC entropy coding
- H.265/HEVC: larger CTU blocks (64×64), improved prediction — same quality at ~50% bitrate of H.264
- Note: MPEG-3 was merged into MPEG-2; MPEG-7/21 are metadata/framework, not video codecs

Text compression

- Text compression aims to reduce storage size without any loss of information.
 - Huffman coding
 - Fewer bits are used to encode frequently occurring characters.
 - More bits are used to encode infrequently occurring characters.
 - A codebook is required.

<i>Symbol</i>	<i>Probability</i>	<i>Code</i>
<i>e</i>	0.8	1
<i>t</i>	0.16	01
<i>x</i>	0.02	001
<i>z</i>	0.02	000

Entropy Coding for Text and Symbols

- Information content: frequent symbols should use shorter codes (Shannon's source coding theorem)
- Fixed-length codes (ASCII 7/8-bit): simple but ignore frequency distribution
- Variable-length codes: Huffman (optimal prefix code for given frequencies), arithmetic coding
- Run-length encoding (RLE): exploit consecutive repetitions — good for fax, simple graphics
- Dictionary coding (LZW): build table of repeated patterns — GIF, Unix compress, early ZIP
- All text methods are lossless — exact reconstruction required

Huffman coding

- The characters are sorted in ascending order of their occurrence frequencies.
- The two characters with the smallest probabilities (leaf nodes) are merged into a parent node whose probability equals the sum of their probabilities.
- The list is reordered to include the remaining characters and the newly created parent node.
- This process is repeated until all characters are merged; the final parent node becomes the root of the Huffman tree.
- Traversing from the root to the leaves (characters), the left branch is labeled '0' and the right branch is labeled '1'.
- The Huffman code for each character is the concatenation of the branch labels along the path from the root to that character.

Text compression

- Run-length encoding
 - replaces a sequence of repeated symbols with a single symbol accompanied by the number of repetitions.

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- A threshold for the number of repetitions must be defined.
- Additional symbols may be introduced.
- Example: “MMMMannnnnnnn” (13 characters) -> “@M4a@n8” (7 characters)

Run-Length Encoding — Extended

- RLE format: (count, symbol) pairs or escape character (@M4 = 4 M's)
- Best case: long runs of identical symbols (black/white fax lines, repeated pixels)
- Worst case: no repeats — RLE adds overhead (e.g., A1B1C1 longer than ABC)
- Hybrid: apply RLE after zigzag scan in JPEG (runs of zero AC coefficients)
- Threshold: only encode as run if repeat count ≥ 3 (avoids expansion on random data)
- Bilingual note: uses RLE on aaabb ccccd aeeee fffff

Exercise — Run-Length Encoding

- Encode using RLE with format: @SymbolCount for runs ≥ 3 , else literal characters.
 - Text: aaabb ccccd aeeee ffff (ignore spaces for counting)
 - (a) Write the RLE-encoded string
 - (b) Compare original length vs encoded length
 - (c) Compute compression percentage
 - (d) Would RLE work well on English prose? Why or why not?
 - (e) Give one multimedia context where RLE is highly effective

Text compression

- LZW coding: A dictionary of character strings (patterns) is used. Codes corresponding to these strings are stored and transmitted instead of codes for individual characters.
- Example with this paragraph:

“Cám ơn đời mỗi sớm mai thức giấc
Ta có thêm ngày nữa để yêu thương
Cám ơn chiều mây âm thầm quên trôi
Làm cơn mưa giăng lối nhỏ em về”

When encoded using LZW, fewer bits may be required compared with conventional ASCII encoding..

LZW Algorithm — Steps

- Initialize dictionary with all single characters (ASCII 0–255)
- Read input; match longest string w in dictionary in the input stream
- Output code for w ; add (previous string + first char of next match) to dictionary
- Repeat until input exhausted — dictionary grows dynamically with patterns
- Decoder rebuilds same dictionary from received codes — no separate table transmission
- Effective on text with repeated words/phrases; GIF uses LZW on pixel color indices

Summary — Multimedia Data Compression

- All MM data types exhibit redundancy — spatial, temporal, spectral, or coding redundancy
- Audio: PCM → predictive/perceptual coding (MPEG) exploits hearing limits
- Image: DCT + quantization + entropy coding (JPEG) exploits visual limits
- Video: adds motion compensation to image techniques (MPEG/H.26x family)
- Text: Huffman, RLE, LZW — always lossless entropy reduction